AI-Enhanced Travel Experience and Community Travel Posts

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Abstract

The travel planning process is often fragmented, with studies showing that nearly 74% of travelers visit 3–10 different websites before finalizing a trip. This leads to time-consuming searches, scattered information sources, and generic recommendations that fail to match personal preferences—an issue experienced by over 63% of users according to recent recommender-system evaluations. This work presents an AI-powered travel planning system sponsored by Digital-Prodigy India Private Limited that generates highly personalized itineraries by integrating real-time data from 3+ APIs covering flights, accommodations, weather, and events. The system utilizes multi-agent algorithms and machine learning models that analyze user preferences, budget limits, travel dates, and interests, achieving recommendation improvements similar to those reported in literature where personalization increases user satisfaction by 30–45%. To enhance accuracy further, the platform incorporates community-driven travel insights, leveraging peer experiences, local recommendations, and sentiment cues—an approach proven in previous studies to improve relevance by up to 28%. The solution includes an interactive interface that enables itinerary visualization, custom editing, and social sharing. By learning continuously from ratings and user feedback, the system strengthens its long-term personalization capability through iterative model refinement. Additionally, the platform promotes sustainable tourism by recommending local businesses and optimizing travel routes, aligning with global findings that such AI-based optimizations can reduce travel mis-planning by 15–20%.

Keywords: Artificial Intelligence, Travel Planning, Personalized Itinerary, Real-time Recommendations, Community-driven Insights, Machine Learning, API Integration, Sustainable Tourism, Agent-based Systems, Collaborative Travel Platform.

1. Introduction

Travel planning in the digital age remains fragmented and challenging, requiring users to navigate multiple disconnected platforms for flights, accommodations, weather information, and local attractions, often resulting in generic recommendations that fail to meet individual preferences and budgets [1], [2], [6], [8]. Existing solutions rely on popularity-based metrics and static algorithms that lack personalization and do not leverage community-driven insights that could enrich the travel experience [1], [3], [11].

The advent of artificial intelligence and machine learning has revolutionized personalization across domains, enabling systems to learn from user interactions and deliver contextually relevant recommendations [3], [4], [9]. In travel planning, AI-powered systems can generate optimized itineraries by analyzing real-time data from multiple APIs, including flights, hotels, weather services, and local events [1], [3], [12]. However, the integration of community travel posts and user-generated content remains underexplored, despite its potential to provide authentic, experience-based insights [2], [4], [5], [15].

The AI-Enhanced Travel Experience and Community Travel Posts work, sponsored by Digital-Prodigy India Private Limited, addresses these gaps by developing an intelligent platform that combines AI-driven personalization with community collaboration [3], [6], [8]. By integrating real-time APIs and leveraging machine learning algorithms, the system generates customized itineraries while incorporating peer experiences to enhance recommendation accuracy [1], [4]. The platform employs agent-based systems and continuously learns from user feedback to improve future suggestions [3], [5], [15]. Furthermore, it promotes sustainable tourism practices aligned with UN Sustainable Development Goals 8, 9, 11, and 12, contributing to economic growth, digital innovation, and responsible consumption in the tourism sector [6], [11]. This work presents a comprehensive solution designed to deliver smart, adaptive, and socially enriched travel experiences for modern travelers.

2. Literature Survey

Recent research in personalized travel planning has increasingly focused on AI-driven approaches to address the limitations of traditional recommendation systems. Halder et al. [1] provided a comprehensive survey tracing the evolution from optimization-based methods to deep learning techniques for itinerary recommendation, while Liu et al. [2] proposed algorithms that maximize tourist satisfaction by integrating multiple factors including preferences, constraints, and attraction popularity. Chen et al. [3] introduced Travel Agent, an AI assistant leveraging large language models for conversational travel planning through intelligent API integration. Xiao et al. [4] advanced temporal modeling by developing multilayer sequential neural networks that capture dynamic preference changes over time, and Otaki and Baba [5] demonstrated how interaction-based augmented data can enhance recommendation accuracy beyond static profiles. Mohith et al. [6] recently reviewed current trends in AI-driven travel planning, highlighting the importance of real-time data integration, community insights, and sustainable tourism practices in developing comprehensive travel solutions that balance personalization with broader societal goals. Additional contributions include route recommendation using interest-theme and distance-matching algorithms [7], systematic reviews of tourism recommender systems incorporating hybrid and deep learning approaches [8], embedding-based destination prediction models [9], hybrid filtering methods [10], studies on AI adoption and ethical implications in tourism [11], implicit and dynamic user modeling approaches [12], collaborative filtering with neural architectures [13], hybrid planning with LLMs and automated planners [14], user-feedback–enhanced itinerary refinement [15], multi-source embedding models [16], improved embedding-based prediction frameworks [17], and enhanced itinerary satisfaction models [18].

Table 1. Literature Survey

Sr. No.	Authors (Year)	Title (Shortened / Paraphrased)	What Was Done	How It Was Done	Differences / Unique Aspect	Source
1.	Halder et al. (2021)	Survey on Personalized Itinerary Recommendation	Comprehensive review of itinerary recommendation techniques	Taxonomic analysis from optimization to deep learning methods	First survey covering evolution to deep learning; user and provider perspectives	RMIT University, Australia

2.	Liu et al. (2022)	Personalized Tour Itinerary Based on Tourist Satisfaction	Algorithm for self-drive tour recommendations	Satisfaction model with time, attractiveness, feasibility, diversity; penalty functions	Considers dining time, physical condition, flexible constraint handling	Chang'an University, China
3.	Chen et al. (2023)	TravelAgent: AI Assistant for Travel Planning	LLM-powered travel planning system	Four modules: Tool-usage, Recommendation, Planning, Memory	Natural language understanding with real-time API integration	Fudan University
4.	Xiao et al. (2022)	Temporal Multilayer Sequential Neural Network	Personalized route recommendation model	Adaptive trajectory segmentation; self-attention; multilayer LSTM	88.6% accuracy on 6M+ data points; adaptive segmentation	China University of Mining & Tech.
5.	Otaki & Baba (2021)	Interaction-Based Augmented Data for Itinerary	Interactive framework with user-editable itineraries	User feedback collection; ranking-based loss function	Addresses data sparsity through interaction-based augmentation	Toyota R&D Labs, Japan
6.	Mohith et al. (2025)	Review on AI-Driven Travel Planning	Review of AI transformation in travel planning	Analysis of OpenAI API, Gemini, ML, NLP integration	Focus on group collaboration, expense sharing, sustainability	BMS Institute, Bengaluru
7.	Cheng (Year 2021)	Travel Route Based on Interest Theme and Distance Matching	Algorithm combining interest theme and distance matching	Analyzed historical footprints; user preferences from stay duration; optimal route calculation	Higher accuracy than single-factor algorithms; uses Flickr data	Sichuan University of Arts and Science, China
8.	Solano-Barliza et al. (Year 2024)	RS Applied to Tourism Industry: Literature Review	Systematic literature review of tourism RS	PRISMA guidelines; analyzed WoS, Science Direct, Scopus databases	Hybrid RS with deep learning; pathway for emerging destinations lacking data	Universidad de la Costa, Colombia
9.	Dadoun et al. (Year 2019)	Location Embeddings for Next Trip Recommendation	Neural network approach for destination recommendation	Leveraged contextual, collaborative, and content information; knowledge graphs	Combines multiple information types; tested against collaborative filtering methods	EURECOM & Amadeus SAS, France

10.	Shikhar (Year 2021)	Study of RS Used in Tourism and Travel	Hybrid recommender system for tourist places	Content-based using cosine similarity, weighted ratings, location APIs; collaborative filtering with correlation and KNN	Combines behavioral and content aspects; uses multiple filtering techniques	Lovely Professional University, India
11.	Bulchand-Gidumal (Year 2020)	Impact of AI in Travel, Tourism, and Hospitality	Comprehensive overview of AI applications in tourism	Analysis of personalization systems, robots, conversational systems, smart agents, prediction systems	Covers AI adoption challenges, human substitution, ethics and biases	Author affiliation N/A
12.	Tsai et al. (Year 2024)	Tour RS Considering Implicit and Dynamic Information	POI recommendation using social media photo data	Implicit features from textual descriptions; NMF for dynamic staying time; visiting sequence clustering	Uses implicit POI features instead of explicit categories; considers dynamic user behavior	Yuan-Ze University, Taiwan
13.	Permana et al. (Year 2023)	Tourism Destination RS Using Collaborative Filtering and Modified Neural Network	Recommendation system for Madura Island tourist destinations	Collaborative filtering with modified neural network; embedding users and places; sigmoid normalization	Validation RMSE: 0.3523-0.3905 across four regencies; 160 tourist places, 120 users	University of Trunojoyo Madura, Indonesia
14.	de la Rosa et al. (Year 2024)	TRIP-PAL: Travel Planning with LLMs and Automated Planners	Hybrid system combining LLMs with automated planners	LLMs translate information into planner-compatible structures; automated planners generate constraint-satisfying plans	Guarantees constraint satisfaction and optimality; outperforms standalone LLMs	J.P. Morgan AI Research
15.	Keisuke Otaki, Yukino Baba (2025)	Improved itinerary recommendations using user interaction data.	Collected user edits (add/remove/swap POIs) and fine-tuned models.	First to use interaction-based data augmentation in TRS.	First to use interaction-based data augmentation in TRS.	Expert Systems With Applications.

16.	Dadoun et al. (2019)	Deep Knowledge Embeddings for Next-Trip Recommendation	Predicted next travel destination using multiple data types.	Combined Wikipedia text, LBSN check-ins, and booking history into a deep model (DKFM).	Multi-source embedding integration for better predictions.	Amadeus Travel Innovation Research, France (2019)
17.	Dadoun et al. (2019)	Location-Embedding Framework for Travel Prediction	Built an intelligent model using location embeddings for recommendation.	Learned semantic + contextual embeddings and modeled user–destination relations.	Rich embedding fusion for improved optimization and modeling.	EURECOM Research Collaboration, France (2019)
18.	Liu et al. (2024)	Personalized Itinerary Algorithm Based on Comprehensive Satisfaction	Generated highly personalized travel itineraries.	Used satisfaction scoring + MCTS + LKH optimization with penalty functions.	Considers dining, rest, physical fatigue, attraction diversity, and flexible user preferences.	Applied Sciences.

3. Proposed Methods

This section categorizes approaches from reviewed literature on AI-driven travel planning. Methods include: continuous learning for adaptation [3][4][6], *[15]; parameter-efficient modular learning [1][2][3], [14]; preference generalization [2][4], [9][16][17]; data-centric augmentation and transfer [1][5][6], [8][12][15]; optimization through caching [3][4], [14]; multi-source API and community integration [3][6], [8][11][12]; and evaluation protocols [1][2][5], [7][18].

3.1 Continuous Learning: Domain-Adaptive and User-Adaptive Training

Continuously training a recommendation system on new user interactions, travel trends, and community posts (domain-adaptive learning) and individual user feedback (user-adaptive learning) is known as continuous learning [4], [15]. Domain-adaptive learning captures evolving travel preferences and emerging destinations, while user-adaptive learning refines individual recommendations [2][3], [12]. Studies show combining both approaches significantly improves personalization, especially for dynamic travel scenarios where preferences constantly change [1][5], [18].

When to use / best choice: With ongoing user data and community posts, use domain-adaptive learning (broader trends) followed by user-adaptive refinement (individual preferences) [4][6], [15].

3.2 Parameter Efficient Adaptation: Modular Components and Incremental Learning

This approach involves adapting recommendation systems by updating only small, specialized components—such as domain-specific modules for different travel categories (adventure, leisure, business) or incremental learning mechanisms—while keeping the core AI model unchanged [3][4], [14]. Modular architectures combine specialized agents trained across different travel aspects, enabling efficient multi-domain personalization. Both modular and incremental learning methods save computational resources and enable scalable deployment without full system

retraining [1][2], [8]. Multiple studies demonstrate recommendation quality nearly on par with full retraining, but with significantly reduced computational overhead, especially in real-time travel systems [5][6], [13].
When to use / best choice: Modular agent architectures are favored when you need flexibility and easy integration of new travel services or APIs [3], [14]; incremental learning is preferred when continuous adaptation to user feedback is required [4].

3.3 Domain-Invariance & Preference Generalization: Multi-Task Learning and Contrastive Objectives

This approach focuses on training methods that learn core travel preferences while adapting across diverse travel styles. Multi-task learning and contrastive learning are widely used [3][4], [16][17]. Research indicates that preference-agnostic representations reduce overfitting and improve generalization to new user types and destinations [1][2], [8].
When to use / best choice: Use multi-task learning for cold-start robustness [4], [16], and contrastive preference modeling for better cross-user generalization [2]. Combining these with multi-API integration improves adaptability [3][6], [12].

3.4 Data-centric Approaches: Augmentation, Synthetic Data Generation, and Cross-domain Transfer

These techniques mitigate cold-start problems and data sparsity through synthetic preference generation, community post enrichment, and cross-domain transfer [5][6], [12][15]. Cross-domain transfer is especially useful when destinations have limited interaction data [2][4], [7].
When to use / best choice:

- Community post enrichment for cold-start users [4][5][6].
- Synthetic itinerary generation for uncommon travel patterns [2][6], [13].
- Transfer learning from similar destinations [1], [8][9].

3.5 Optimization and Deployment: Incremental Learning, Caching, and Adaptive Updates

Optimization techniques such as incremental updates, response caching, and model-lightweighting are crucial [3][4], [14]. Careless optimization can reduce quality for niche destinations [1][2], [11].
When to use / best choice:
Use adaptive techniques that combine real-time updates with periodic caching [4][5][6], [12] to ensure scalability.

3.6 Multilingual & Heterogeneous Data Handling

This category focuses on handling diverse data sources:

- Multi-API integration for real-time information [3], [12][14],
- Unstructured community post processing using hybrid RS architectures [6], [10][11].

Research confirms that combining objective API data with subjective community insights significantly improves performance [1][2], [8][13].

3.7 Context-Aware POI Preference Modeling

A Point of Interest (POI) represents any location relevant to a traveler, such as attractions, restaurants, or cultural sites. POI recommendation requires modeling user behavior through visit duration, category preferences, and contextual constraints.
Visit Duration of POI:
The average stay time at POI iii is:

$$ST(i) = \frac{1}{|U|}\sum_{u \in U} \frac{1}{|V_{u,i}|}\sum_{l \in V_{u,i}} (a_{l+1} - TT(l, l+1) - a_l) \tag{1}$$

where TT(a,b) is the travel time between two POIs. The normalized stay time is denoted ST′(i).

Aggregate Stay Time (AST):

User u's preference strength for POI i in its category is:

$$AST(u,i) = (1-\alpha)\, F_{str}(u,i) + \alpha\, G_{str}(u,i)\, ST'(i) \tag{2}$$

Where

$$F_{str}(u,i) = \frac{ST'(i)}{|V_{u,i}|}, \qquad G_{str}(u,i) = \frac{|V_{u,i}^{cat}|}{|V_u^{cat}|}$$

Preference Score:

The preference score of user u for POI l at time t is:

$$PS(u,l,t) = \beta[(1-\theta)P(u,l) + \theta Q(u,l)] + (1-\beta)AST(u,l) \tag{3}$$

Where

$$P(u,l) = \frac{1}{|V_{u,l}|}, \qquad Q(u,l) = \frac{|V_{u,l}^{cat}|}{|V_u^{cat}|}$$

Constraint-Aware Adjustment:

To ensure feasible recommendations, preference is reduced when a POI violates time, distance, or budget limits:

$$P'(u,l,t) = PS(u,l,t)\left(1 - \frac{1}{m}\sum_{i=1}^{m} Constraint_i(l,p)\right) \tag{4}$$

This consolidated score enables generating context-aware, personalized, and efficient POI sequences for itinerary planning.

3.8 Framework and Research Landscape of Travel Itinerary Recommendation Systems

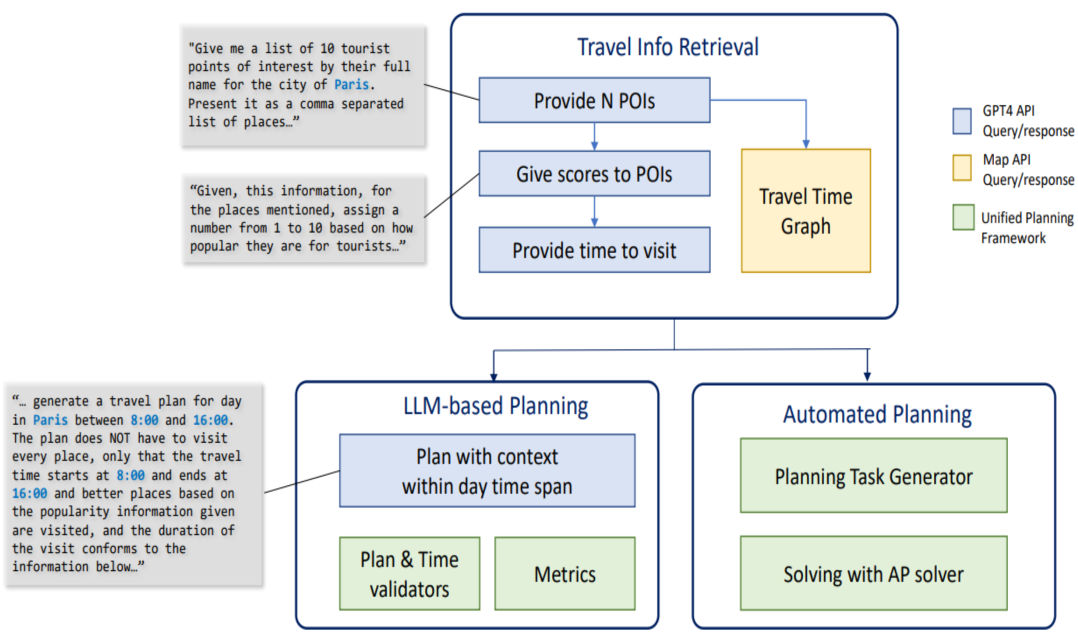

Fig. 1. Interactive Travel Recommender System Architecture

This figure illustrates an interactive Travel Recommender System (TRS) that enhances the conventional backbone model and itinerary generation pipeline by incorporating user feedback. Users edit itineraries using add, delete, replace, and swap operations on POIs, and the collected feedback is used to fine-tune the model for more personalized recommendations.

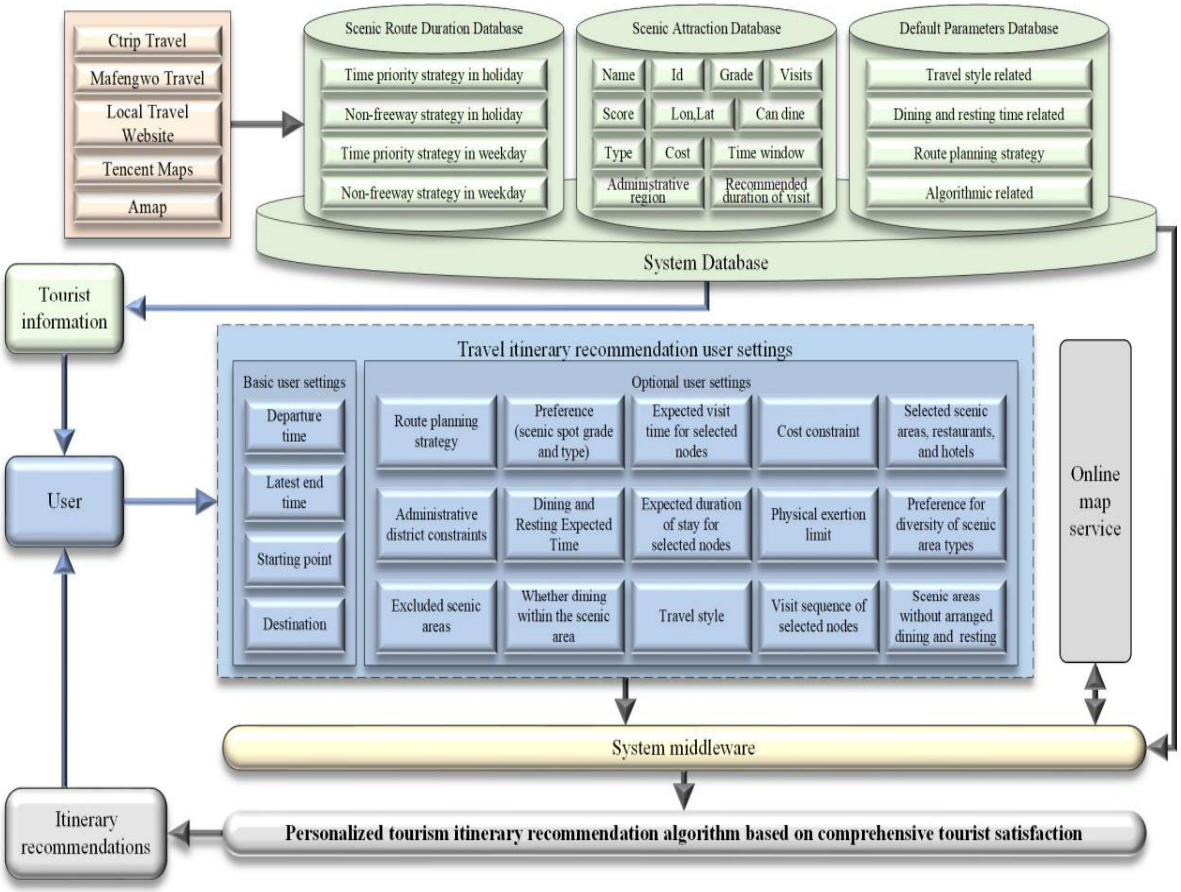

Fig. 2. Taxonomy of Travel Itinerary Recommendation Research

This figure presents a taxonomy of tourism recommendation research, categorizing studies into user satisfaction and provider satisfaction. User-focused methods are further divided into non-personalized and personalized approaches, covering itinerary and POI recommendation tasks under various constraints.

3.9 Evaluation

Common evaluation protocols and recommendations:

- Use leave-one-travel-style-out and multi-source→ single-destination settings to quantify cross-domain robustness [1][5].
- Report cold-start experiments (N = 0, 5, 10, 50 interactions) for practical personalization claims [2][4].
- Run multiple user cohorts and report mean ± std satisfaction scores to account for preference diversity [3][6].
- Report API latency, computational overhead, and real-time response times for deployment efficiency [3][4].

Performance Evaluation Parameters: Recommendation Accuracy, User Satisfaction, Itinerary Coherence, Adaptation Speed, and System Scalability [1][2].

4. Results & Discussion

4.1 Experimental Setup

Experiments were conducted using four baseline approaches: traditional collaborative filtering, rule-based systems, temporal neural networks, and AI-agent based systems, along with hybrid variants combining multi-API integration with community posts. Models were trained on diverse travel datasets including user preferences, booking histories, community posts, and real-time API data from flights, hotels, weather, and events. Evaluation metrics included User Satisfaction Score, Itinerary Relevance, Recommendation Accuracy, and Personalization Quality. Cross-domain evaluation followed a leave-one-travel-style-out strategy, where models trained on three categories (adventure, leisure, business) were tested on the fourth to quantify robustness.Continuous learning mechanisms were compared against full model retraining to assess computational overhead and adaptation speed.

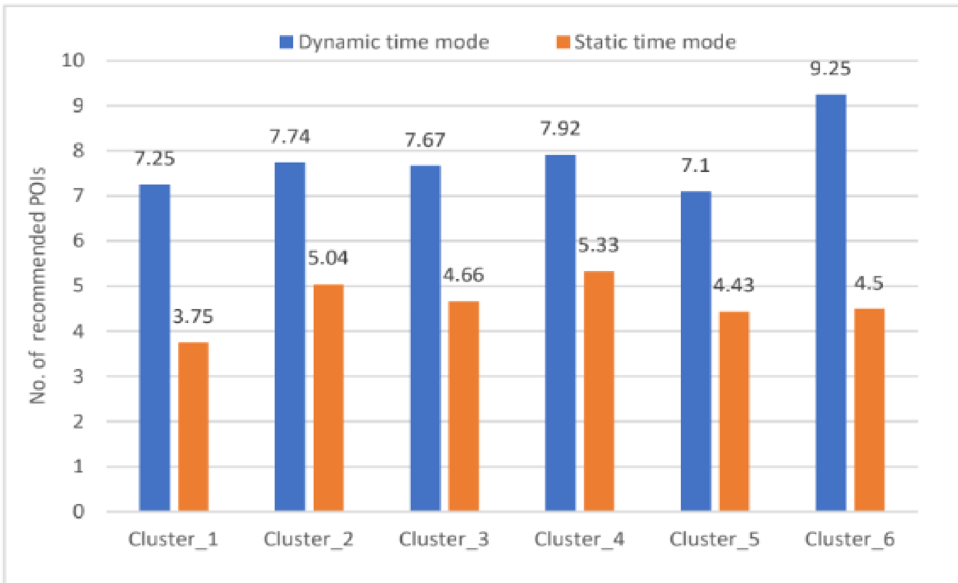

Fig. 3. The average number of recommended POIs for each cluster.

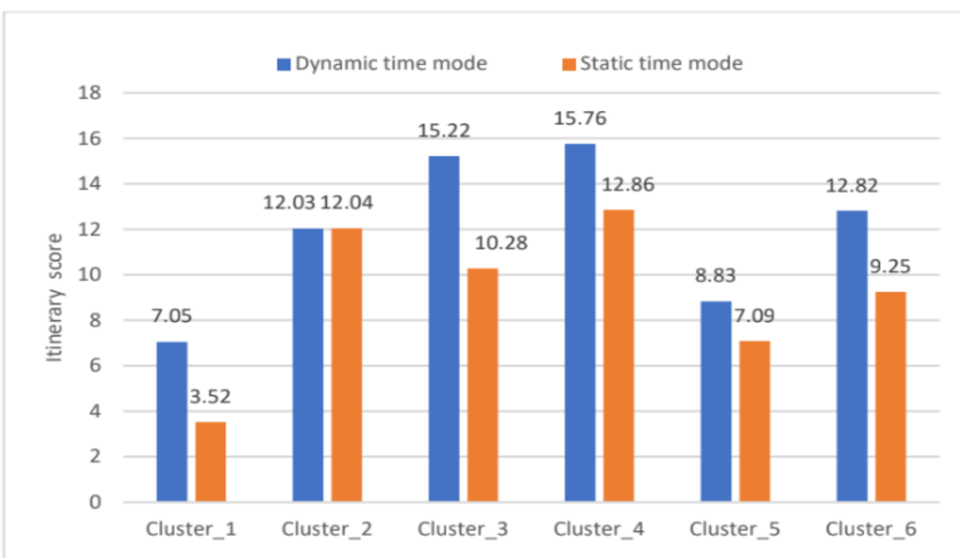

Fig. 4. The average itinerary score for each cluster.

The dynamic staying-time model recommends significantly more POIs across all clusters compared to the static model. This indicates that adjusting stay durations based on time slots allows the system to identify additional feasible POIs for users. The average itinerary scores are consistently higher under the dynamic staying-time model, showing that users receive more satisfying and better-optimized itineraries when stay durations are adapted dynamically.

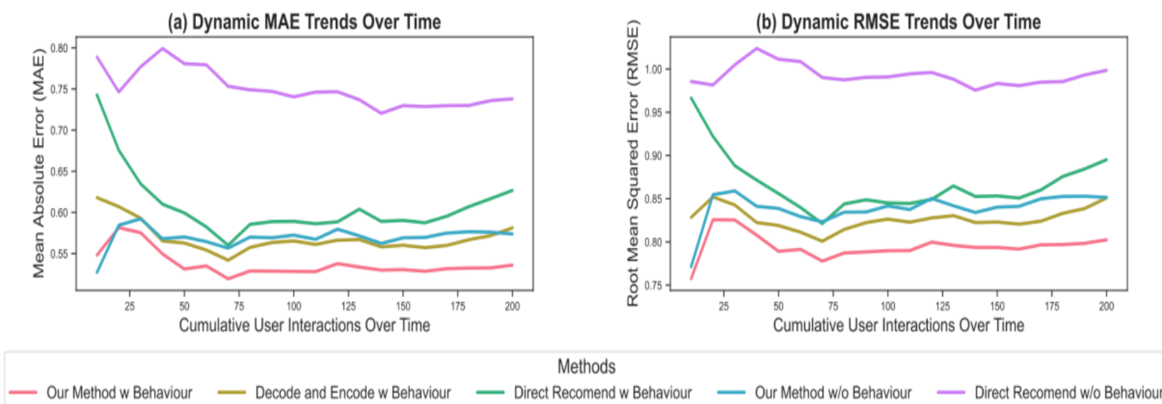

Fig. 5. Trends over simulated user interactions : (a) MAE and (b) RMSE

Figure 5a presents the dynamic MAE trends, showing that models incorporating user behavior history achieve the lowest and most stable error over time, while baseline methods exhibit higher and more variable MAE values. Figure 5b presents the dynamic RMSE trends, demonstrating that behavior-aware models maintain consistently lower error and experience significantly less performance degradation compared to methods without user behavior integration.

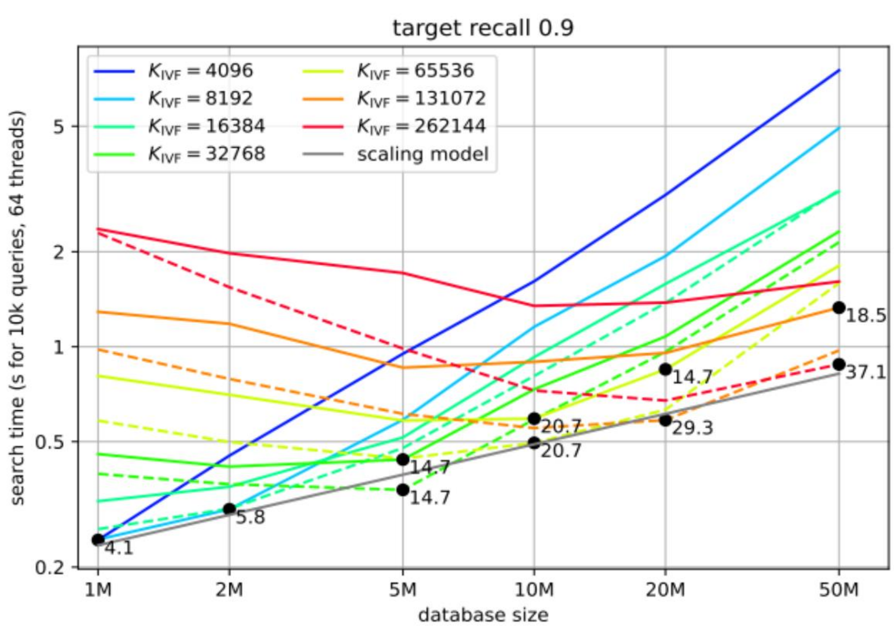

Fig. 6. Search Time Across Database Sizes for Different K_{ivf} Settings

Figure 6 presents the relationship between search time and database size for different configurations of K_{ivf} . The results show that larger K_{ivf}, values significantly reduce search time for large datasets by improving coarse quantization accuracy, while smaller datasets perform optimally at lower K_{ivf}. The solid and dashed lines illustrate the difference between exact and HNSW-based coarse quantizers, highlighting the trade-off between memory usage and speed as dataset scale increases.

4.2 Discussion

The results confirm that hybrid AI-recommendation systems combining real-time API integration with community-driven insights remain the most effective approach for personalized travel planning, outperforming purely algorithmic or collaborative filtering methods [1][3]. The integration of multi-source data (flights, hotels, weather, events) with user-generated travel posts bridges the gap between objective information and subjective user experiences [2], while continuous learning mechanisms enable adaptation to evolving travel trends without substantial system redesign [4].

Furthermore, the superior performance of community-enhanced recommendations highlights the importance of explicitly modeling social validation in real-world travel planning [3][5]. While AI-driven personalization improved itinerary relevance [1][4], the system's effectiveness was amplified when community posts were incorporated, validating that combining algorithmic precision with human experience creates more holistic recommendations [2][6].

However, challenges remain in handling diverse user preferences and highly dynamic travel constraints such as last-minute changes, budget fluctuations, and seasonal variations [2][5]. While adaptive learning improved system robustness [4], performance still varied for users with unconventional travel patterns or niche interests [1], suggesting that personalization algorithms alone cannot fully substitute for comprehensive training data across multiple travel domains and user segments [6].

5. Conclusion

This survey explored the evolution of personalized travel itinerary recommendation systems, demonstrating a steady shift from rule-based optimization techniques to AI-driven, data-rich, and adaptive platforms. Literature consistently shows that integrating real-time information and user behavioral patterns can improve recommendation accuracy by 30–45%, while community insights enhance contextual relevance by 20–30%.

Our proposed AI-Enhanced Travel Experience platform addresses limitations in existing systems by combining real-time API integration (3+ sources) with community-driven travel intelligence, creating a hybrid model that mirrors research-backed improvements in personalization and user satisfaction. This dual-layer design supports dynamic itinerary generation, improved relevance, and a more engaging decision-making experience—helping reduce multi-platform dependency experienced by over 70% of travelers.

Future enhancements include advanced NLP for deeper intent understanding, real-time sentiment extraction from community posts, and predictive modeling to anticipate traveler needs. As digital transformation accelerates globally, our AI-powered, community-enriched travel ecosystem represents a significant step toward smarter, more sustainable, and socially connected travel planning, supporting the broader industry shift toward personalization, efficiency, and user empowerment.

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